

Coursework Submission Coversheet

Please use this coversheet for all coursework assessment submissions.

You should copy the completed coversheet into your assessment document as the first page and save as one single document for submission.

If you require full guidance on coursework submission, please consult the Code of Practice on Assessment (available in your Programme Organisation in Minerva). In particular:

- See section 3.3d for the full policy on penalties for exceeding assessment limits (e.g. word count).
- See section 3.3e for the full policy on penalties for late submission of coursework.

To complete this coversheet, please provide the following information:

Module code: OCOM5201M

Module title: Knowledge Representation and Reasoning

Assessment title: Assessment 2: Ontology, Prolog, Modal Logic

Your 9-digit Student ID number: 201889137

Word count or minute count: 316

Submission date: 15 Jun, 2025

Please note that by submitting this piece of work you are agreeing to the University's [Declaration of Academic Integrity](#).

1. Overview

The ontology models the structure of sandwiches that captures the whole (`Sandwich`), its parts (`BreadSlice`, `LayerOfFilling`, `Filling`), and six variants (`SingleDeckerSandwich`, `DoubleDeckerSandwich`, `OpenSandwich`, `ToastedSandwich`, `Panini`, `ClubSandwich`).

I also created five object properties describe how these elements are related: `hasBreadSlice` that links a sandwich to each slice of bread; `hasLayer` links a sandwich to each vertical layer of filling; `hasFillingIngredient` links a layer to any filling substance (`hasMeatFilling` and `hasVegetableFilling`)

2. Justification

For classes, `Sandwich` is placed directly under `owl:Thing`; everything that is a sandwich inherits from it.

Then, `SingleDeckerSandwich`, `DoubleDeckerSandwich`, `OpenSandwich`, `ToastedSandwich`, `Panini`, `ClubSandwich` are declared subclasses of `Sandwich` because each one is a sandwich with additional constraints.

`BreadSlice`, `LayerOfFilling`, and `Filling` remain siblings of `Sandwich` because they are parts of a sandwich, not specialised sandwiches themselves.

For properties, `hasBreadSlice` and `hasLayer` are general relations from a sandwich to its components, so they have no parent other than `owl:topObjectProperty`.

`hasFillingIngredient` is a relation from a layer to any filling. `hasMeatFilling` and `hasVegetableFilling` are declared sub-properties of `hasFillingIngredient`.

3. Axiom

For `SingleDeckerSandwich` the ontology contains `hasBreadSlice` exactly 2 `BreadSlice` and `hasLayer` exactly 1 `LayerOfFilling`. This mean every `SingleDeckerSandwich` must have exactly two bread slices and exactly one deck of filling.

For `DoubleDeckerSandwich` the ontology contains `hasBreadSlice` exactly 3 `BreadSlice` and `hasLayer` exactly 2 `LayerOfFilling`. This means every `DoubleDeckerSandwich` has exactly three slices and two decks of filling.

4. Advantages and disadvantages

The advantages of this ontology focus on the structural and compositional aspects of a sandwich, like how many slices of bread it has, how many filling layers, and the distinction between meat and vegetable ingredients. This will help it to keep a clear and focused scope.

The disadvantage of this structure is that there is little further subdivision of the variants. All variants are below `Sandwich`, it can be harder to capture more detailed groupings or shared behaviour among subsets of sandwiches.